Databases: establishing firmer foundations

Objectives:

In this lab you will

- Explore the structure of a SQL query
- Investigate the structure of a sample relational database
- Practice some DQL queries to retrieve data from our sample database
- Use DDL statements to create and alter objects within our sample database
- Leverage DML statements to alter records within our sample database

Motivation (Why are we doing this?):

This lab will allow us to not only set you on the right path with this module, but also give you an opportunity to assess and build on your current knowledge and skill level as it pertains to the basics of using database systems so that we can feed this into improving our materials and teaching approach in later weeks ☺

Understanding how to effectively use Database systems is a key skill for anyone involved in Data Engineering / Machine Learning Operations, our assessment will ask us to not only access Data from a database but to also create our own database structure for storing data which will be used for:

- Creating our machine learning models
- Analysing and reporting on important data which may be used to generate insight for our organisation / company.

Important

In these early labs the focus will be on asking you to explore and familiarise yourself with the basics of the environment. This is not only to give you the ability to dive into some practical tasks right from the start, but also so that we can better gauge your current understanding and skill level when it comes to utilising some of the tools and technologies that will be leveraged throughout the module. The labs are constructed in such a way that we **expect** you to ask further questions in class, to help develop a deeper understanding. If you find yourself getting stuck, or feeling that you don't fully understand an element of the task, don't panic! ☺ Just make sure you let us know as soon as possible.

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Select

What is it?

The select statement is used to retrieve records from a database, you are selecting the attributes that you want to retrieve for each record as the DBMS scans through the database.

Why might I use it?

SELECT statements are one of the simplest queries we can construct and are used as the basis for most queries that involve retrieving records from one or more tables.

For instance, we may wish to retrieve all the record tuples that are within our Customers table.

How can it be done?

Example

```
SELECT *
FROM Customer;
```

Exercise

Construct 5 SQL statements that use this technique, try to create a variety of queries using this technique.

When constructing your statements you should include a summary of:

- What your statement or query does
  - A summary of the query in terms of **what is being done** within the SQL statement itself from the **perspective of the database**
    - This would discuss the query from the perspective of what it is doing in with the data objects (tables, attributes, etc) within the database from a more technical perspective
  - A summary of the query in terms of the **real-life information it is trying to extract / change** from the **perspective of a user in a business setting**
    - This would discuss the query from the **perspective of what it is doing in terms of the actual useful business information / data it is trying to find** from the **perspective of a user trying to extract / change further information about an aspect of the business**
- Why your statement does this

- An explanation of **why your statement might be used in the real world** for the purpose of **extracting / changing information relevant to aspects of the business**

- How your statement operates
  - Explain each **clause** in your **statement** and discuss how each of those clauses/keywords contribute to constructing the overall statement

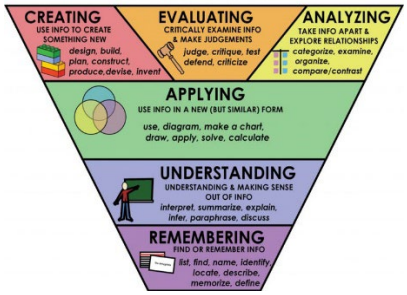

Figure 1 Taxonomy of Learning. Image courtesy of Rawia Inaim / Kwantlen Polytechnic University

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Note
If you are unsure about how to approach attempting aspects of this exercise, discuss with the tutor as soon as possible – I am here to help! 😊

Table Aliases

What is it?

SQL table aliases are used to give a table a temporary name that it can be referred to by. Tables are created by following the **FROM** clause with the **AS** keyword, which will create an alias for the duration of that query, outside of the specific execution of that query the alias will not exist.

The upshot of this is that aliases are almost like local variables in programming, which can be used to refer to your tables, they only exist within a given executing statement (in other words we cannot reference the alias in another statement) and the alias will be destroyed at the end of that statement execution (again somewhat like a local variable in programming)

Why might I use it?

Table aliases are often used to make table names easier to refer to within a SQL statement. For example, we may wish to use the ALIAS 'c' instead of Customer as shorthand to refer to the customer table.

This has somewhat more limited usage in queries that only involve one table, but in larger queries that may involve multiple tables aliases are very useful to allow us to keep track of which table we are referring to at any given point via our shorthand alias value.

How can it be done?

Example

```
SELECT c.*
FROM Customer AS c;
```

Exercise

Construct 5 SQL statements that use this technique, try to create a variety of queries using this technique.

When constructing your statements you should include a summary of:

- What your statement or query does
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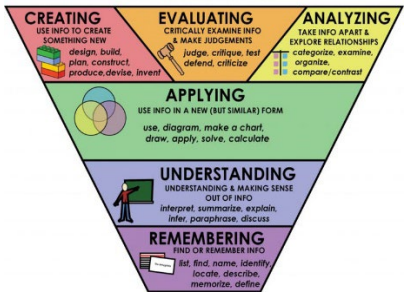

Figure 2 Taxonomy of Learning. Image courtesy of Rawia Inaim / Kwantlen Polytechnic University

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Qualified select: Qualifying attributes (vertical qualification)

What is it?

A vertically qualified select statement allows us to specify individual attributes (columns) that we wish to extract from a given query result. It is 'vertically qualified' because attributes (columns) are (at least conceptually) thought of as the vertical labels in our results set, therefore it specifies ways to extract or 'slice' or result data "vertically".

Why might I use it?

How can it be done?

We use an appropriately constructed SQL select statement to tell the DBMS to perform this task.

We can assume (for the time being) that the database engine will understand that it needs to first **access the data object** representing the specific **table** in question it will **scan through each record** for the entire table, but only note down the data within the **specifically selected attributes** for that record.

For instance, if we take the following table

Attribute1	**Attribute2**	**Attribute3**	**Attribute4**
Record 1 Attribute1	Record 1 Attribute2	Record 1 Attribute3	Record 1 Attribute4
Record 2 Attribute1	Record 2 Attribute2	Record 2 Attribute3	Record 2 Attribute4

Figure 3 ExampleTable

And execute this SELECT statement

```
SELECT Attribute1, Attribute3
FROM ExampleTable;
```

The DBMS will understand that it only needs to extract Attribute1 and Attribute3 for each record in the result set.

Attribute1	**Attribute2**	**Attribute3**	**Attribute4**
Record 1 Attribute1	Record 1 Attribute2	Record 1 Attribute3	Record 1 Attribute4
Record 2 Attribute1	Record 2 Attribute2	Record 2 Attribute3	Record 2 Attribute4

Figure 4 Data that is read and extracted by the database engine is highlighted in blue

So it will return the following result

Attribute1	**Attribute3**
Record 1 Attribute1	Record 1 Attribute3

Record 2 Attribute1	Record 2 Attribute3

Figure 5 Data that the database engine has read and extracted is highlighted in blue

Example

```
SELECT Id, CompanyName, ContactName, Country, City
FROM Customer;
```

Exercise

Construct 5 SQL statements that use this technique, try to create a variety of queries using this technique.

When constructing your statements you should include a summary of:

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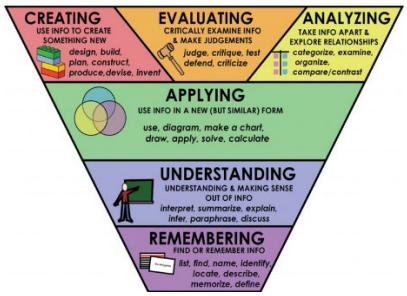

Figure 6 Taxonomy of Learning. Image courtesy of Rawia Inaim / Kwantlen Polytechnic University

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Note
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Attribute aliases

What is it?

Attribute aliases allow us to take a specific attribute that we are retrieving from our intermediate result set and give it a new temporary name / identifier in the result set that is returned.

Why might I use it?

Attribute aliases allow us to give more meaningful / readable names to particular attributes that are being retrieved. It is particularly useful when using tables where the meaning of the attribute names is not entirely clear, thus by given the attributes an alias we can give additional context as to their meaning within the set of results retrieved.

Attribute aliases are also useful when performing JOINs between two or more tables, as they allow us to rename attributes within the context of our result set rather than within the context of an individual table, a very simple example of this is that you could have an ID attribute on the Customer table as well as an ID attribute on the Order table without aliasing it will not be clear within the result set of a join between the two tables which ID belongs to which relation / table.

How can it be done?

Example

```
SELECT Id AS 'Customer ID', CompanyName AS 'Name of Company',  Address
AS 'Company Address', Country, City
FROM Customer;
```

Exercise

Construct 5 SQL statements that use this technique, try to create a variety of queries using this technique.

When constructing your statements you should include a summary of:

- What your statement or query does
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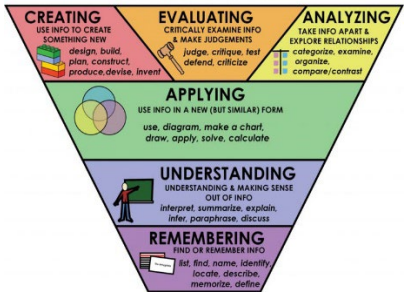

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Qualified select: Qualifying records (horizontal qualification)

What is it?

Qualified select allows us to tell the DBMS to filter our results so that only specific records are retrieved from the database.

Why might I use it?

We use horizontal qualification when we only want records that meet a specific condition (predicate). For instance, we may want to only retrieve Customer details for our UK customers.

How can it be done?

We use the WHERE clause to allow us to filter the result set retrieved from a table based on a specific condition provided.

As well as SELECT statements we can also use the WHERE clause to qualify UPDATE and DELETE operations.

Example

```
SELECT *
FROM Customer
WHERE Country = 'UK';
```

Exercise

Construct 5 SQL statements that use this technique, try to create a variety of queries using this technique.

When constructing your statements you should include a summary of:

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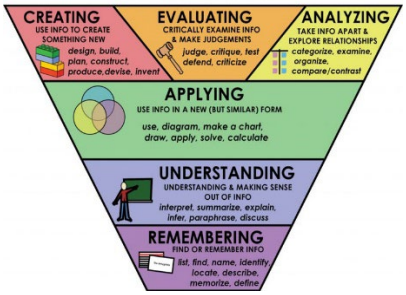

Figure 8 Taxonomy of Learning. Image courtesy of Rawia Inaim / Kwantlen Polytechnic University

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Note
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Aggregate functions: Count

What is it?

The Count aggregate function allows us to ask the DBMS to calculate how many records are provided in our set of results for a given query.

Why might I use it?

We can use the Count function when we want to count the amount of complete records retrieved from the database. For instance, we may want to count the amount of Customers that we have.

How can it be done?

Example

```
SELECT Count(*)
FROM Customer;
```

Exercise

Construct 5 SQL statements that use this technique, try to create a variety of queries using this technique.

When constructing your statements you should include a summary of:

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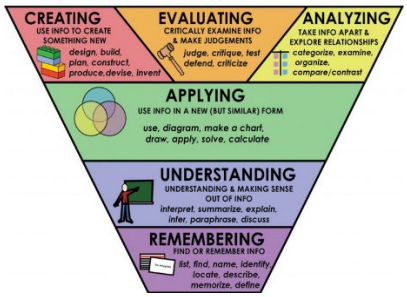

Figure 9 Taxonomy of Learning. Image courtesy of Rawia Inaim / Kwantlen Polytechnic University

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Unqualified Join / Cross Join: The Cartesian Product

What is it?

A way of pairing the data within each records for one table with the data for each record in another table.

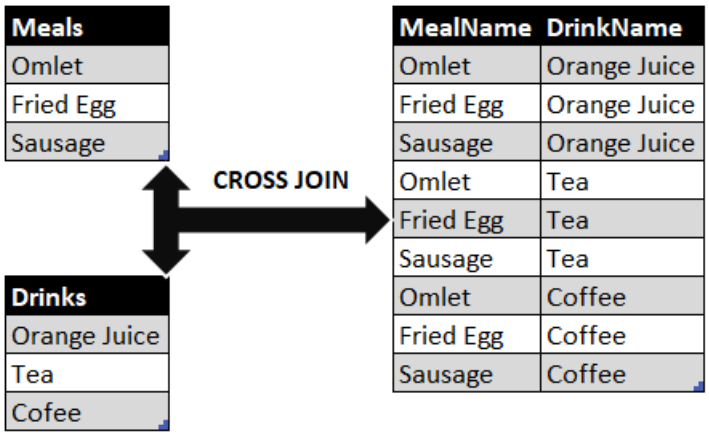

Figure 10 Performing a CROSS JOIN on two example tables to retrieve the cartesian product of the two tables

Why might I use it?

Because they join all the data for each record in one table with all the data for each record in another table CROSS JOIN operations are usually limited in their practical usefulness. One of their major uses is as an intermediary step to demonstrate how INNER JOIN operations construct their result sets.

Another major use of the CROSS JOIN operation is to test the performance of a given DBMS (and the database it is managing) when handling queries that need to combine data from multiple tables.

How can it be done?

There are two major ways of performing a CROSS JOIN in SQL. One is simply by performing a SELECT operation on two tables, without a WHERE clause which attempts to match relevant data from one table with another.

The other method is to simply perform a SELECT operation on one table and use the CROSS JOIN keyword to introduce the second table that should be included in the result set.

Example

```
SELECT o.Id AS 'Order ID',
       o.CustomerID AS 'Customer ID on Order table',
       c.Id AS 'Customer ID on Customer Table',
       c.CompanyName AS 'Company Name'
FROM 'Order' AS o, Customer AS c;
```

Figure 11 Selecting records from two tables will automatically produce an unqualified join

```
SELECT o.Id AS 'Order ID',
       o.CustomerID AS 'Customer ID on Order table',
       c.Id AS 'Customer ID on Customer Table',
       c.CompanyName AS 'Company Name'
FROM 'Order' AS o
CROSS JOIN Customer AS c;
```

Figure 12 An unqualified join can also be performed using the CROSS JOIN keyword

Exercise

Construct 5 SQL statements that use this technique, try to create a variety of queries using this technique.

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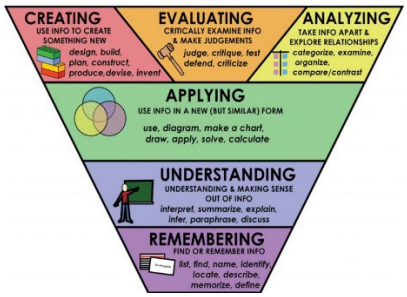

Figure 13 Taxonomy of Learning. Image courtesy of Rawia Inaim / Kwantlen Polytechnic University

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Note
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Inner Join

What is it?

An inner join allows you to select records from both tables that **match** based on a specific criterion (called a join condition) provided by one or more **predicate** conditions.

Why might I use it?

One of the main benefits of database systems is the ability to find related data records in different locations (e.g. tables). An inner join allows us to better identify which records in a given table also have related data in another table, the relationship under investigation is what has been specified in the join condition.

For instance, say we wanted to find all the orders that have been placed for each customer, we would join the Order table with the Customer table and use the join criteria that for each record in the order table we only retrieve the company record(s) for the company that placed that order (based on the company ID).

How can it be done?

Conceptually the inner join can be thought of as a horizontally qualified cross join, in other words it states which records from the cartesian product of two tables to extract as part of the result set of records.

Example

```
SELECT o.Id AS 'Order ID',
       o.CustomerID AS 'Customer ID on Order table',
       c.Id AS 'Customer ID on Customer Table',
       c.CompanyName AS 'Company Name'
FROM 'Order' AS o, Customer AS c
WHERE o.CustomerID = c.Id;
```

Figure 14 Inner Join as a qualified select query of multiple tables

```
SELECT o.Id AS 'Order ID',
       o.CustomerID AS 'Customer ID on Order table',
       c.Id AS 'Customer ID on Customer Table',
       c.CompanyName AS 'Company Name'
FROM 'Order' AS o
CROSS JOIN Customer AS c
WHERE o.CustomerID = c.Id;
```

Figure 15 Inner join as a qualified select query of the cartesian product of two tables

```
SELECT o.Id AS 'Order ID',
       o.CustomerID AS 'Customer ID on Order table',
       c.Id AS 'Customer ID on Customer Table',
       c.CompanyName AS 'Company Name'
FROM 'Order' AS o
INNER JOIN Customer AS c
ON o.CustomerID = c.Id;
```

Figure 16 Inner join using the explicit Inner Join syntax

Food for thought

Taking our knowledge of programming languages and the fact that tools like compilers and interpreters can automatically analyse our code to identify optimisations that can be applied to our programs, which of the above examples of inner join do you believe would likely be the easiest for the database engine to identify optimisations for?

Food for thought

Can you turn the explicit inner join example found above into a cross join by modifying the join condition only? If so, how would you do it and why does / doesn't it work?

Exercise

Construct 5 SQL statements that use this technique, try to create a variety of queries using this technique.

When constructing your statements you should include a summary of:

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  - Explain each **clause** in your **statement** and discuss how each of those clauses/keywords contribute to constructing the overall statement

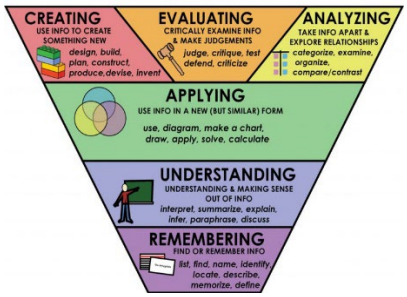

Figure 17 Taxonomy of Learning. Image courtesy of Rawia Inaim / Kwantlen Polytechnic University

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Note
If you are unsure about how to approach attempting aspects of this exercise, discuss with the tutor as soon as possible – I am here to help! 😊

Left Join

What is it?

The **LEFT JOIN** returns **all those records that match the join criterion** between the first **"Left"** table 'Customer' and the second **"Right"** table 'Order'

AND

It also **returns any records in the "Left table" that do not meet the join criteria**.

Why might I use it?

Let's say that we wanted to retrieve data which identifies the specific orders that have been placed by a company AND we also wanted to include those companies which have not placed any orders at all.

How can it be done?

We use the LEFT JOIN keyword to specify that we want to produce a set of results that joins records from two tables together, based on our join criterion. But the LEFT JOIN will mean that it will also include records in the first table that do not have related data in the second table (in other words it will still include companies who haven't placed an order).

Example

Let's first insert a record into our Customer table that we know will have no orders assigned to it.

```
INSERT INTO Customer
VALUES ('NOORDER', 'DoesntOrderInc', 'Jaqen Hghar',
          'Faceless Man', 'House of Black and White', 'Braavos',
            NULL, 'GOT5', 'Essos', NULL, NULL);
```

Then let's write our Left Join query

```
SELECT c.CompanyName, o.ID
FROM Customer AS c
LEFT JOIN 'Order' AS o
ON c.ID = o.CustomerID;
```

If we look through the result set we would be able to find a record for 'Doesn't Order Inc' where their Order ID is "None" or NULL, this is because the **LEFT JOIN** has returned **all those records that match the join criterion** between the first **"Left"** table 'Customer' and the second **"Right"** table 'Order'

AND

It also **returns any records in the "Left table" that do not meet the join criteria**.

In this way we can produce a result set that contains a list of all Customers and the Orders they have placed that also includes Customers who have not placed an Order yet.

Food for thought

Compare the above LEFT JOIN example with a similarly structured inner join, what do you notice about the set of company results?

Exercise

Construct 5 SQL statements that use this technique, try to create a variety of queries using this technique.

When constructing your statements you should include a summary of:

- What your statement or query does
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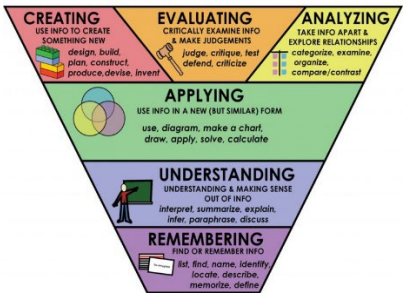

Figure 18 Taxonomy of Learning. Image courtesy of Rawia Inaim / Kwantlen Polytechnic University

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Note
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Group by

What is it?

The Group By statement allows us to group records that have the same values together, to provide a summary of attributes for that group.

Why might I use it?

We tend to use the group by statement when we have multiple records that are related to a particular value.

For instance, we may wish to work out the amount of orders that have been placed by a particular customer to do this we would group our result set by an attribute that allows us to uniquely identify each company we would then count the amount of records within the database that are Orders related to that Company (this may involve a Join condition if the information is contained in multiple tables

How can it be done?

We use the Group By statement to identify the attribute(s) within our result set that we wish to group together. For instance, if we wanted to group together records for each Company we would select an attribute that can be used to identify each Company and tell the DBMS to group the results based on that attribute.

If we wanted to group together records of our customers for each Country in order to find the amount of Customers in each country we would select an attribute that can be used to identify the Country of each customer (this could be a name, an ID, a country code, etc) and tell the DBMS to group the results based on that attribute.

Group By is commonly used with one or more aggregate functions to allow an aggregated summary to be calculated for each group of records that the result set has been divided into.

Example

```
SELECT c.CompanyName, Count(o.ID) AS 'Amount of Orders'
FROM Customer AS c
LEFT JOIN 'Order' AS o
ON c.ID = o.CustomerID
GROUP BY c.CompanyName;
```

```
SELECT Country, COUNT(ID) AS 'Amount of Customers'
FROM Customer
GROUP BY Country;
```

Exercise

Construct 5 SQL statements that use this technique, try to create a variety of queries using this technique.

When constructing your statements you should include a summary of:

- What your statement or query does
  - A summary of the query in terms of **what is being done** within the SQL statement itself from the **perspective of the database**
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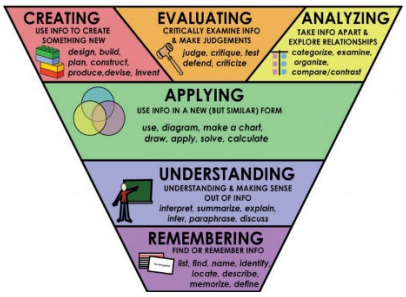

Figure 19 Taxonomy of Learning. Image courtesy of Rawia Inaim / Kwantlen Polytechnic University

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Order by

What is it?

Allows us to tell the DBMS to sort our result set in a particular alphanumeric order based on some attribute (column)

Why might I use it?

We may wish to look at the set of all orders for each Customer, to ensure that the orders for a Customer are all together in our result set we might choose to Order our results based on a unique identifier for the Customer.

How can it be done?

We use the Order By clause to tell the database what attribute it must use to sort the records. Essentially an initial result set of relevant records will be retrieved by the database which will then have some sorting algorithm applied to the results, the sorting algorithm will use the specific attribute chosen to decide the sorting order of the results set retrieved. In this way we go from the usual notion of having an unordered collection of results to an ordered collection of results.

Example

Let's ensure that we have inserted the record for Doesn'tOrderInc into the Customer table.

```
INSERT INTO Customer
VALUES ('NOORDER', 'DoesntOrderInc', 'Jaqen Hghar',
          'Faceless Man', 'House of Black and White', 'Braavos',
            NULL, 'GOT5', 'Essos', NULL, NULL);
```

Then let's perform our previous left join to retrieve the Orders associated with each company, where is DoesntOrderInc in the result set?

```
SELECT c.CompanyName, o.ID AS 'Order ID'
FROM Customer AS c
LEFT JOIN 'Order' AS o
ON c.ID = o.CustomerID;
```

It is likely near the bottom of the result set, this is because records in a database table are not stored in any specific order by default as a table is just a an unordered collection of records.

Now let's order our query by company name, this means that the result set of records will be sorted into alphanumeric order by the Company Name attribute.

```
SELECT c.CompanyName, o.ID AS 'Order ID'
FROM Customer AS c
LEFT JOIN 'Order' AS o
ON c.ID = o.CustomerID
ORDER BY c.CompanyName;
```

Food for thought

Why did I believe that it was likely that the DoesntOrderInc record was near the bottom of the result set? What does this tell us about how records might be stored within our database?

Exercise

Construct 5 SQL statements that use this technique, try to create a variety of queries using this technique.

When constructing your statements you should include a summary of:

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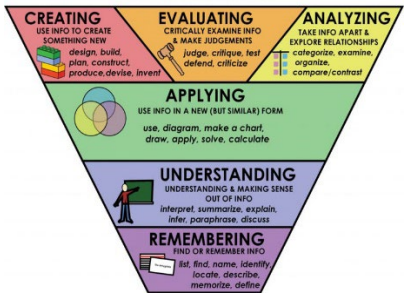

Figure 20 Taxonomy of Learning. Image courtesy of Rawia Inaim / Kwantlen Polytechnic University

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Nested Queries

What is it?

A query within another query, the inner nested or 'sub' query is used to retrieve intermediate results that are then used by the outer query.

Why might I use it?

We tend to use subqueries for data retrieval tasks that require more complicated logic / decision making than would be allowed in a simpler query and that would either not be feasible to perform via a join OR would likely make the resulting query too complicated if joins were used instead. For instance, if we wanted to retrieve those Customers that are not in the same Country as one of our Suppliers, we might first write a query to find the countries our suppliers are from and then use that as part of a condition to only extract those Customer records who are not in one of those Countries

How can it be done?

We can place a nested query in parentheses in most places where an expression would be expected within our SQL statements.

Example

Let's write a nested query to:

1. Get a set of records specifying the Countries that our Suppliers are in
2. Get a set of records of our Customers that are not in the same Country as any of our suppliers
3. Order the results by Country

```
SELECT *
FROM Customer
WHERE Country
        NOT IN (SELECT Country
            FROM Supplier)
ORDER BY Country;
```

Notice, that the inner "nested" query "SELECT Country FROM Supplier" must execute first before the outer query can be executed, because the results of the outer query depend on the data provided from the inner query.

Food for thought

How could we modify the above query to allow us to retrieve all those Customers that are in the same country as one of our suppliers?

Exercise

Construct 5 SQL statements that use this technique, try to create a variety of queries using this technique.

When constructing your statements you should include a summary of:

- What your statement or query does
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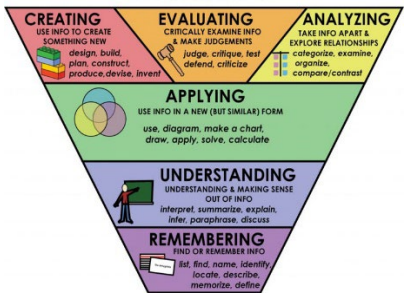

Figure 21 Taxonomy of Learning. Image courtesy of Rawia Inaim / Kwantlen Polytechnic University

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Note
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Attention!
The below statements will mutate (change) the structure of the database, make sure you have a backup copy of the database before trying to make any of these changes to it

Creating tables

What is it?

A method of defining the structure of tables within the database in terms of the table name, attributes and the constraints placed upon them.

Why might I use it?

When you want to create a Table of data 😊

Whenever you need to model a collection of related data records for a given entity

Essentially when you want a part of your database that collates together a number of records which all represent values of the same overall object of interest (Customer, Order, Product, etc).

How can it be done?

We use the CREATE TABLE statement to tell the DBMS we wish to define a new table structure, we also provide the attribute names, attribute types and various additional constraints we may wish to place on the table.

Example

```
CREATE TABLE ContactInfo(
  ContactID INT,
  EmailAddress VARCHAR(255),
  LandlineNumber VARCHAR(255),
  MobileNumber VARCHAR(255)
);
```

Food for thought

At the moment our table definition is very simple and while the data model does provide some basic types for each attribute in the it table does not appropriately capture other important constraints that we may wish to place upon it to ensure that individual records within our table contain valid ContactInfo.

Think about what records would be classed as valid members of the ContactInfo table defined above.Try to produce 5 instances of records that contain valid real world contact info and 5 instances of records that would be valid based on our definition, but would not be valid real world records of Contact Information.

What kind of additional constraints might we want to place on the attributes within the table to prevent records that do not contain valid real world contact information from being classed as valid by the database?

Exercise

Construct 5 SQL statements that use this technique, try to create a variety of queries using this technique.

When constructing your statements you should include a summary of:

- What your statement or query does
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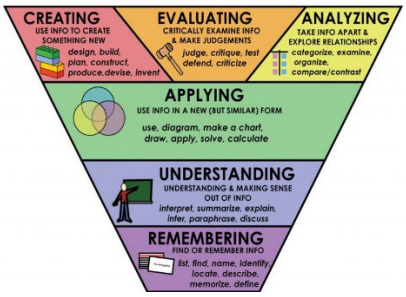

Figure 22 Taxonomy of Learning. Image courtesy of Rawia Inaim / Kwantlen Polytechnic University

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Note
If you are unsure about how to approach attempting aspects of this exercise, discuss with the tutor as soon as possible – I am here to help! 😊

Dropping tables

What is it?

Gives us the ability to remove tables from the database.

Why might I use it?

Whenever we need to remove a given entity within the database. This may be because we created a temporary table for development / debugging purposes or because we are refactoring (redesigning) the database to have a different structure.

How can it be done?

We use the DROP TABLE statement along with the table name.

Example

```
DROP TABLE ContactInfo;
```

Exercise

Construct 5 SQL statements that use this technique, try to create a variety of queries using this technique.

When constructing your statements you should include a summary of:

- What your statement or query does
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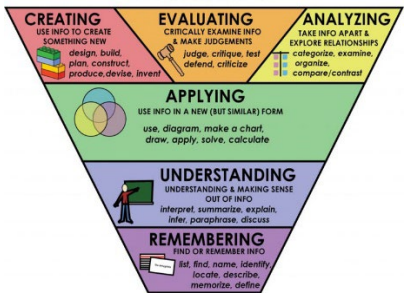

Figure 23 Taxonomy of Learning. Image courtesy of Rawia Inaim / Kwantlen Polytechnic University

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Note
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Creating indexes

What is it?

A method of creating an additional data structure in our database that can help to speed up the process of search for particular records based on a combination of one or more attributes.

Why might I use it?

We will discuss this more later in the module 😊

Suffice to say, we use indices wherever we believe we can get a sufficient performance gain for queries that use a given table's attributes.

How can it be done?

We simply use the CREATE INDEX statement, give the index a name and then identify the table and attributes that we will build the index data structure for.

Example

```
CREATE INDEX idx_customer_country
ON Customer(Country)
```

Exercise

Construct 5 SQL statements that use this technique, try to create a variety of queries using this technique.

When constructing your statements you should include a summary of:

- What your statement or query does
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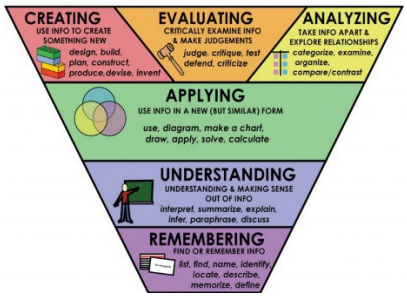

Figure 24 Taxonomy of Learning. Image courtesy of Rawia Inaim / Kwantlen Polytechnic University

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Note
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Dropping indexes

What is it?

Gives us the ability to tell the DBMS to remove an index on a given set of attributes for a table.

Why might I use it?

Inappropriate use of indexing can actually slow down search queries in some situations, they also require additional datastorage and processing that needs to be updated and maintained by the DBMS.

How can it be done?

We can use the DROP INDEX statement and then identify the name of the index that we wish to remove from the database.

Example

```
DROP INDEX idx_customer_country
```

Creating views

What is it?

Views allow us to essentially create intermediate result sets for a given query. In this way instead of sending queries to a particular table, we can instead query a view, to retrieve our desired results

Why might I use it?

When we want to have particular predefined ways of requesting data from the database, that already has already had some additional structure or constraints applied to it than would usually be found within the underlying table. For example, we may wish to have a view of the Customer table that only includes those customers from a particular country.

How can it be done?

We use the CREATE VIEW statement then we identify the name we wish to give the view and the query that will be used to structure the result set that will be presented as a view to the user.

Example

```
CREATE VIEW view_customer_uk AS
SELECT *
FROM Customer
WHERE Country = 'UK';
```

Exercise

Construct 5 SQL statements that use this technique, try to create a variety of queries using this technique.

When constructing your statements you should include a summary of:

- What your statement or query does
    - A summary of the query in terms of **what is being done** within the SQL statement itself from the **perspective of the database**
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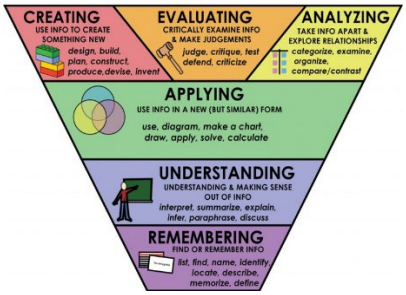

Figure 25 Taxonomy of Learning. Image courtesy of Rawia Inaim / Kwantlen Polytechnic University

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Note
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Dropping views

What is it?

A method of removing views from our database.

Why might I use it?

Sometimes views may be unnecessary and have lost their usefulness, in this case we can tell the DBMS remove these now less useful views from the database

How can it be done?

Simply tell the DBMS to DROP the view and identify which view needs to be removed.

Example

```
DROP VIEW view_customer_uk;
```

Exercise

Construct 5 SQL statements that use this technique, try to create a variety of queries using this technique.

When constructing your statements you should include a summary of:

- What your statement or query does
  - A summary of the query in terms of **what is being done** within the SQL statement itself from the **perspective of the database**
    - This would discuss the query from the perspective of what it is doing in with the data objects (tables, attributes, etc) within the database from a more technical perspective
  - A summary of the query in terms of the **real-life information it is trying to extract / change** from the **perspective of a user in a business setting**
    - This would discuss the query from the **perspective of what it is doing in terms of the actual useful business information / data it is trying to find** from the **perspective of a user trying to extract / change further information about an aspect of the business**
- Why your statement does this
  - An explanation of **why your statement might be used in the real world** for the purpose of **extracting / changing information relevant to aspects of the business**
- How your statement operates
  - Explain each **clause** in your **statement** and discuss how each of those clauses/keywords contribute to constructing the overall statement

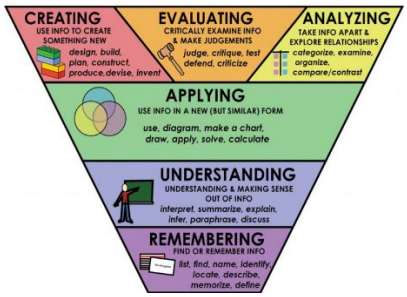

Figure 26 Taxonomy of Learning. Image courtesy of Rawia Inaim / Kwantlen Polytechnic University

This exercise should be seen as an opportunity to consolidate your thought process and high-level understanding of the basics of this SQL technique. With this exercise you should focus on strengthening the lower levels of your learning stack: **remembering**, **understanding** and **applying** the **knowledge** and **skills** gained.

Note
If you are unsure about how to approach attempting aspects of this exercise, discuss with the tutor as soon as possible – I am here to help! 😊

SQL: Manipulating data within our database using DML statement

Attention!
The below statements will mutate (change) values (such as records) within the database, make sure you have a backup copy of the database before trying to make any of these changes to it

Inserting data into a table

What is it?

A way of inserting records into a table.

Why might I use it?

Whenever you need to add new records into the database.

How can it be done?

We use the INSERT INTO statement along with the table name and give the VALUE of the record to be inserted as a tuple.

Example

We've already seen an example of this previously 😊

```
INSERT INTO Customer
VALUES ('NOORDER', 'DoesntOrderInc', 'Jaqen Hghar',
          'Faceless Man', 'House of Black and White', 'Braavos',
            NULL, 'GOT5', 'Essos', NULL, NULL);
```

Exercise

Construct 5 SQL statements that use this technique, try to create a variety of queries using this technique.

When constructing your statements you should include a summary of:

- What your statement or query does
  - A summary of the query in terms of **what is being done** within the SQL statement itself from the **perspective of the database**
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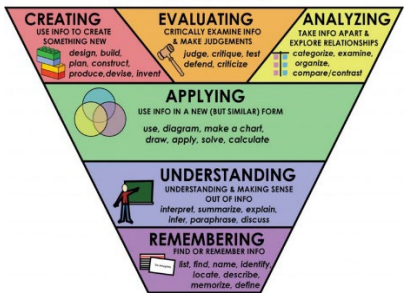

Figure 27 Taxonomy of Learning. Image courtesy of Rawia Inaim / Kwantlen Polytechnic University

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Note
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Updating a record in our table

What is it?

A means of updating an existing record in a table.

Why might I use it?

Whenever we want to update one or more attribute values within a given record for a table, but we do not wish to insert a new record.

How can it be done?

We use the UPDATE statement, identify the table we are updating we then use the SET keyword to identify what attributes to change the value for, the WHERE clause allows us to individually identify the records to update.

Example

```
UPDATE Customer
SET ContactName = 'A Man who is no one'
WHERE CompanyName='DoesntOrderInc';
```

Exercise

Construct 5 SQL statements that use this technique, try to create a variety of queries using this technique.

When constructing your statements you should include a summary of:

- What your statement or query does
  - A summary of the query in terms of **what is being done** within the SQL statement itself from the **perspective of the database**
    - This would discuss the query from the perspective of what it is doing in with the data objects (tables, attributes, etc) within the database from a more technical perspective
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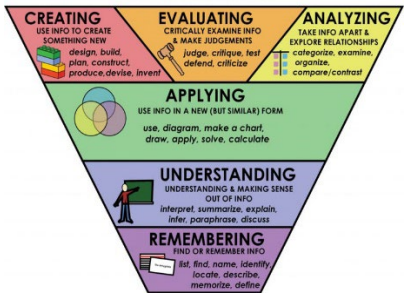

Figure 28 Taxonomy of Learning. Image courtesy of Rawia Inaim / Kwantlen Polytechnic University

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Note
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Deleting a record in our table

What is it?

A method of deleting records within the database

Why might I use it?

Whenever we want to remove a given record for a table

How can it be done?

We use a DELETE statement and identify what table we want to delete FROM, then we use the WHERE clause to identify specific rows.

Example

```
DELETE
FROM CUSTOMER
WHERE Id = 'NOORDER';
```

Exercise

Construct 5 SQL statements that use this technique, try to create a variety of queries using this technique.

When constructing your statements you should include a summary of:

- What your statement or query does
  - A summary of the query in terms of **what is being done** within the SQL statement itself from the **perspective of the database**
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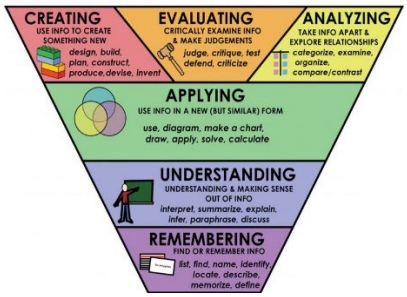

Figure 29 Taxonomy of Learning. Image courtesy of Rawia Inaim / Kwantlen Polytechnic University

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Note
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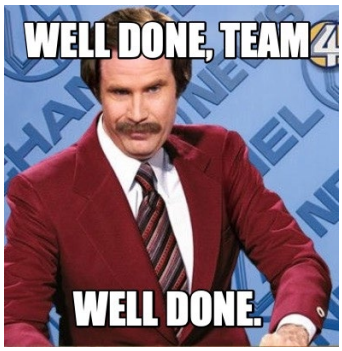
Extension task



Finished early? Excellent! ☺

Use this opportunity to discuss with your lab tutor what you have understood so far, and ask them to assign some tasks that you could attempt to further refine your skills and knowledge.

Here are some suggested sample items you could investigate and then discuss with your tutor:

- Consolidating some of our developing Relational Database knowledge
  - https://amrtechuniverse.com/how-i-met-your-database-and-mainly-relational-using-sql
  -
- Existing Data Analyses of the Northwind Database
  - https://amrtechuniverse.com/how-to-data-analysis-with-sql
  - https://bonnynichol.medium.com/northwind-database-analysis-f106304d1743
  - https://joewebb11.github.io/northwinds_business_analysis
  - https://github.com/ndleah/northwind
  - https://nathanwyand.com/2019/03/27/digging-for-business-insights-in-the-northwind-database/
  - https://medium.com/@mattheweparker/northwind-dataset-exploration-9066425ebaa5